Sri Sivasubramaniya Nadar College of Engineering, Chennai

(An autonomous Institution affiliated to Anna University)

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Academic year	2025-2026 Even	Batch	2023-2027
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Experiment 3: Regression Analysis using Linear and Regularized Models

1.Aim and Objective:

The aim of this experiment is to implement Linear Regression and regularized regression models such as Ridge, Lasso, and Elastic Net for predicting the loan sanction amount and to evaluate their performance using appropriate regression metrics. The experiment also focuses on performing data preprocessing, visualizing feature–target relationships, tuning model hyperparameters through cross-validation, and analyzing overfitting, underfitting, and the bias–variance trade-off to identify the most suitable model.

2. Dataset Description

The dataset is a real-world loan application dataset containing both numerical and categorical attributes related to applicants and property information.

Key attributes include:

- Age, Income, Credit Score
- Profession, Employment Type, Location
- Property Price and Property Type
- Number of Dependents

Target Variable:

- Loan Sanction Amount (USD)

3. Preprocessing Steps:

1. **Removal of Identifier Columns**
   Customer ID, Name, and Property ID were dropped as they do not contribute to prediction.

2. **Handling Missing Values**

   - Numerical features → median imputation
   - Categorical features → most frequent value

3. **Feature Scaling**
   Numerical features were standardized using StandardScaler.

4. **Dataset Splitting**
   The data was divided into training (70%), validation (15%), and testing (15%) sets.

4. Implementation Details

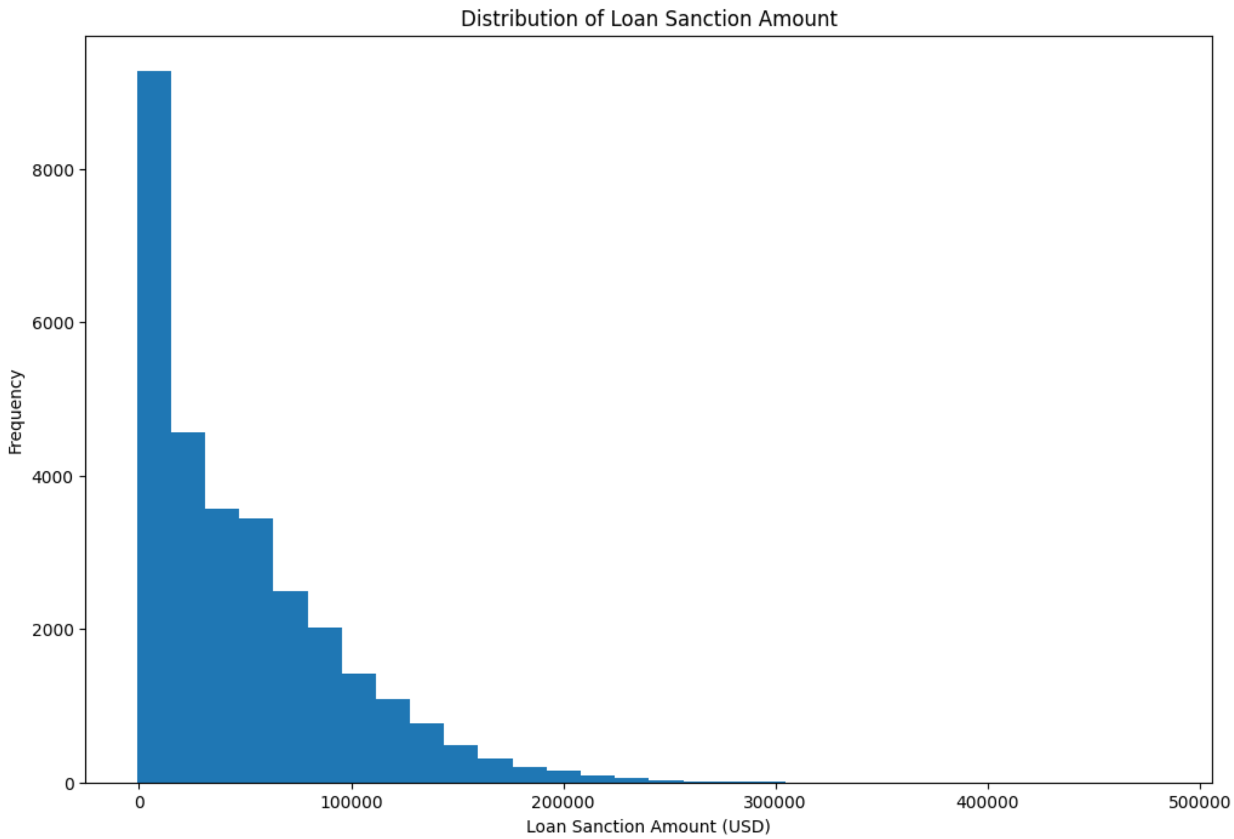
The implementation included:

- Implemented Linear Regression, Ridge Regression, Lasso Regression, and Elastic Net models to predict the loan sanction amount.
- Performed hyperparameter tuning for Ridge, Lasso, and Elastic Net using GridSearchCV with 5-fold cross-validation to determine optimal regularization parameters.
- Evaluated model performance using cross-validated R^2 scores and regression metrics including MAE, MSE, RMSE, and R^2 on validation and test datasets.
- Visualized model behavior using predicted versus actual value plots, residual plots, training versus validation error curves, learning curves, and coefficient comparison bar charts.
- Compared baseline and regularized models to analyze generalization ability and the effect of regularization strength.
- Selected the best-performing model based on test set performance and overall stability.

5. Visualizations

5.1. Target variable distribution plot

Figure 1: Target variable distribution plot



5.2. Feature vs target scatter plots

Figure 2: Numerical features Distributions

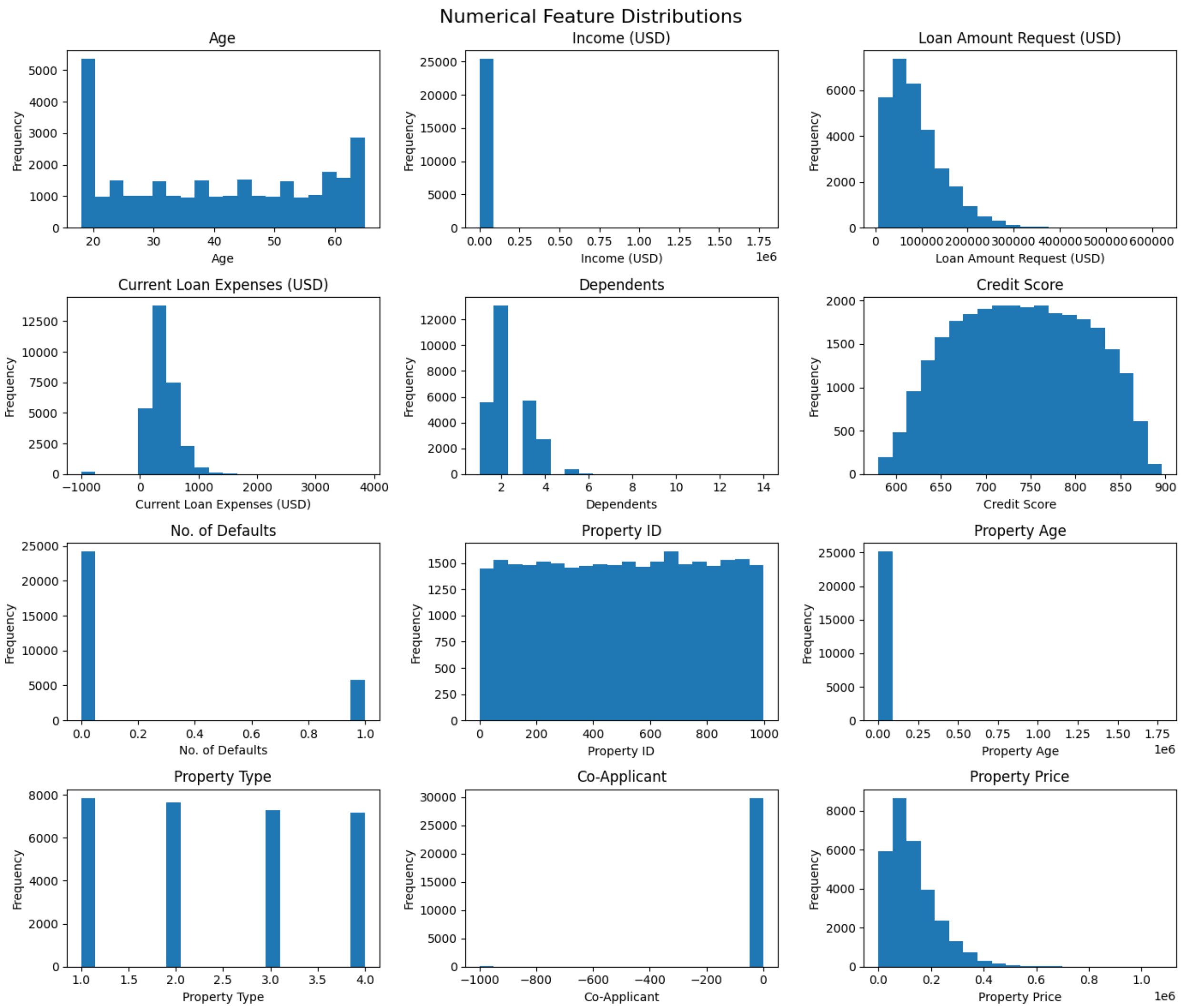


Figure 3: Numerical features vs Target variable Scatterplot

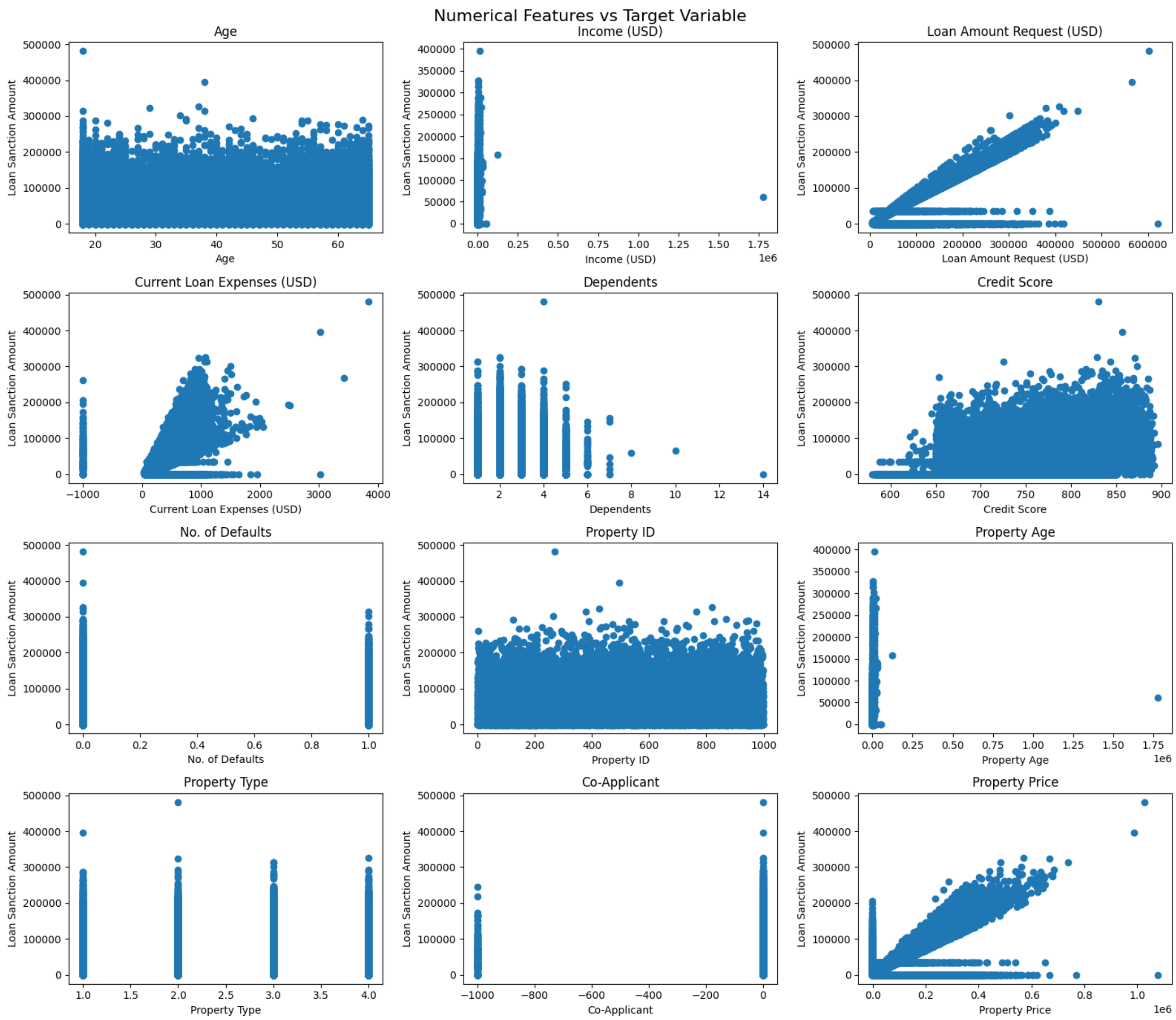


Figure 4: Boxplots for Outlier detection

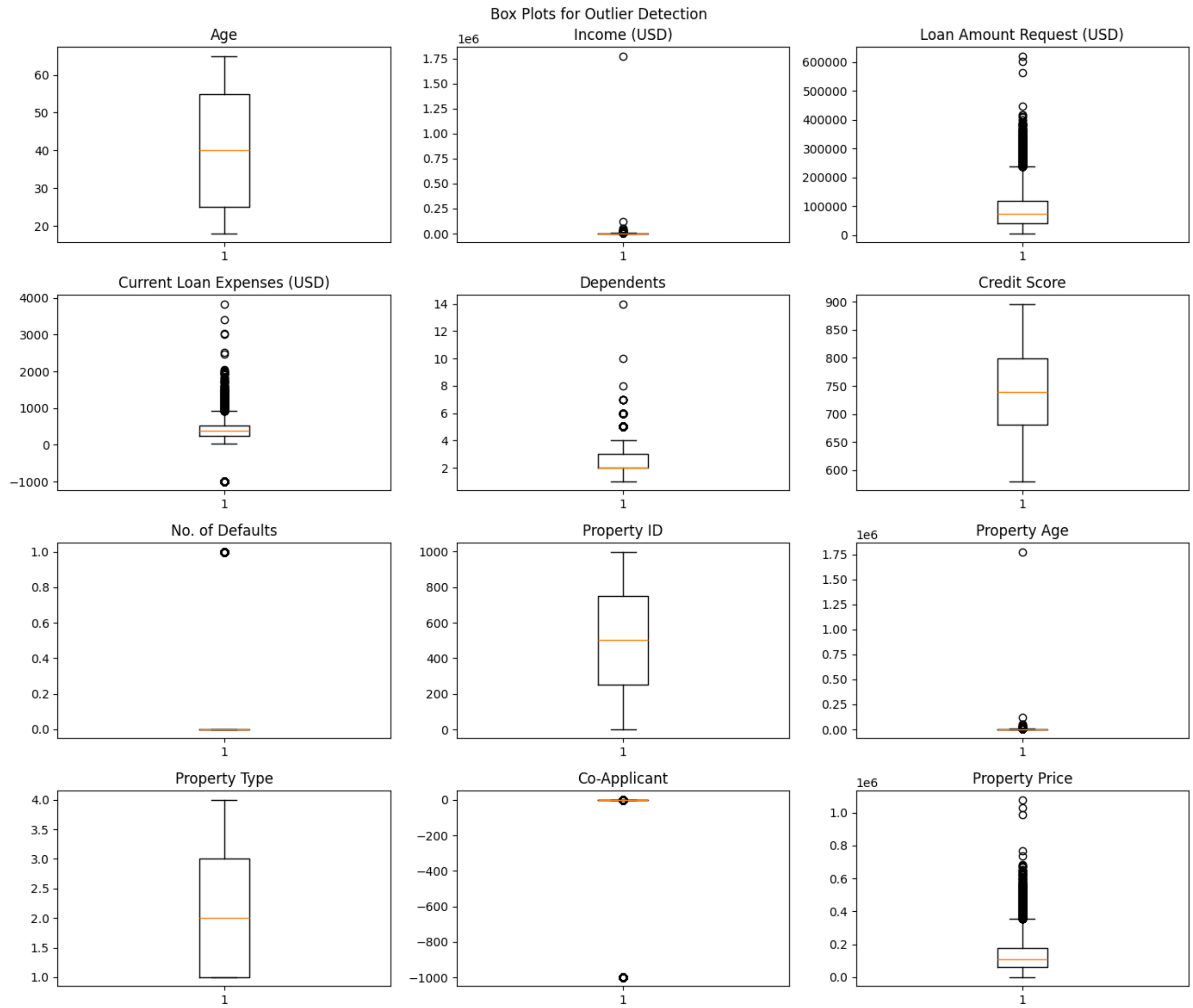


Figure 5: Correlation Heatmap

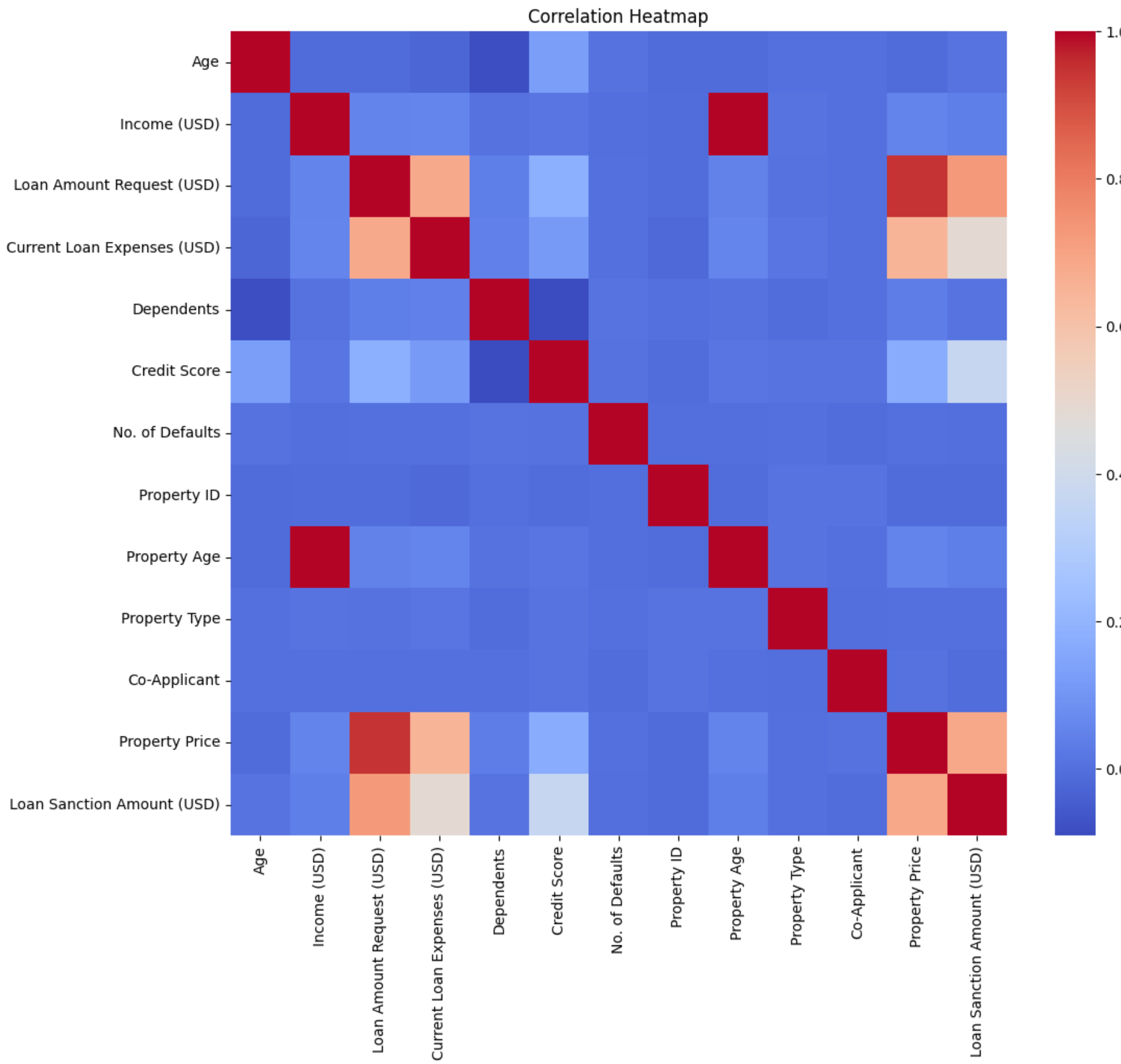
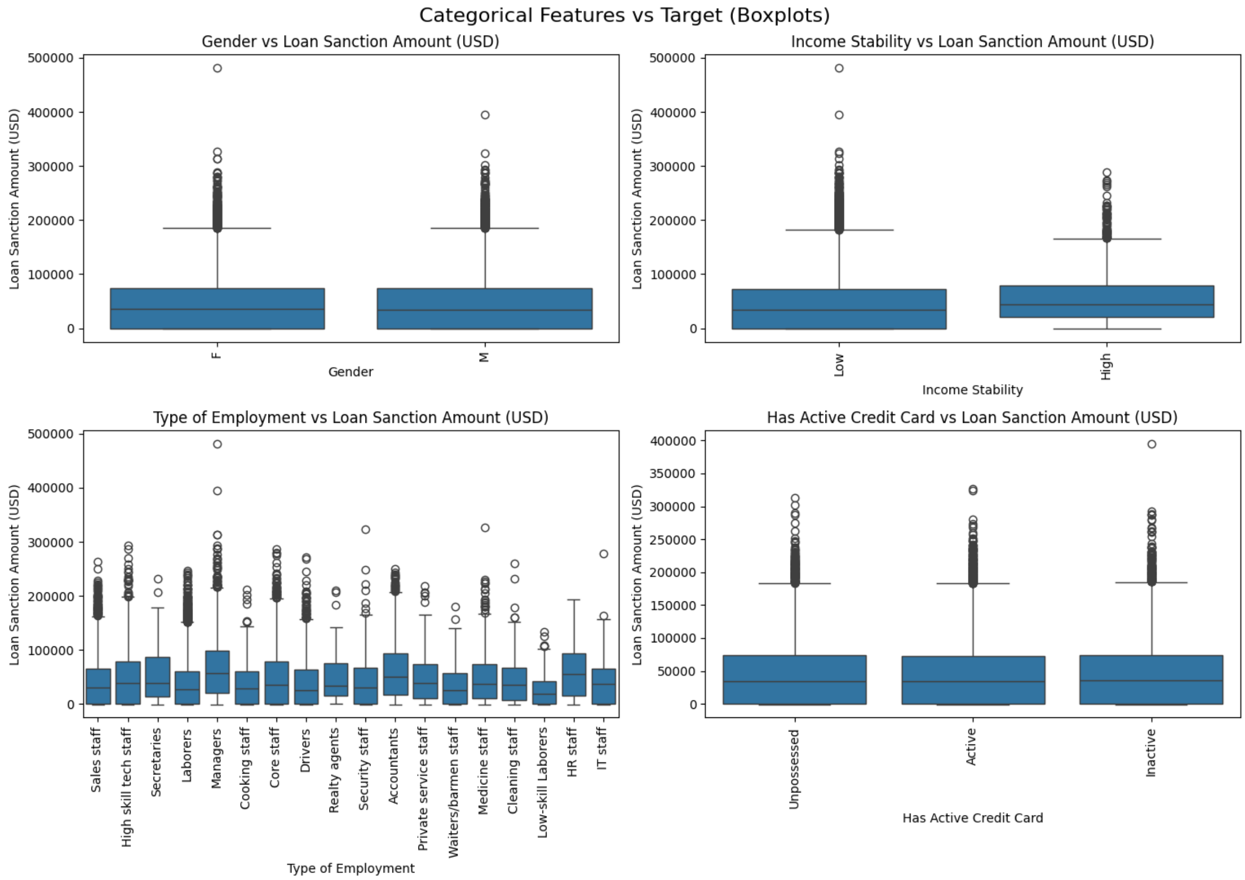


Figure 6: Categorical features vs Target in Boxplots



5.3.Predicted vs actual values plots (for all models)

Figure 9: Predicted vs actual values plot - Linear

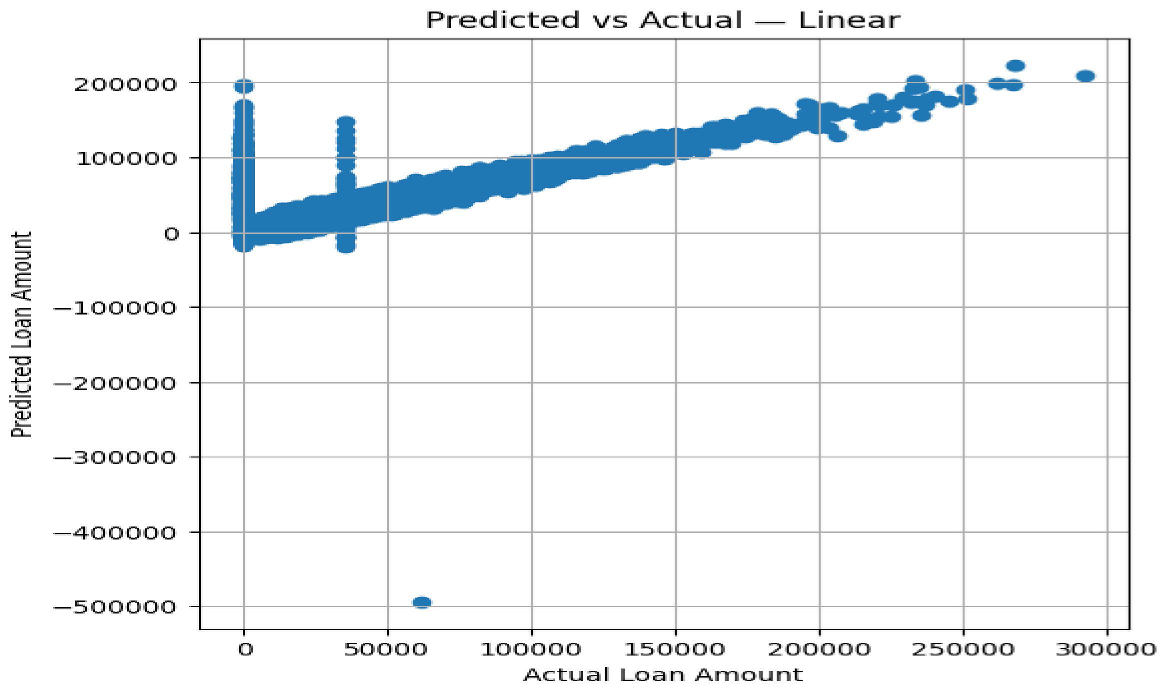


Figure 10: Predicted vs actual values plot - Ridge

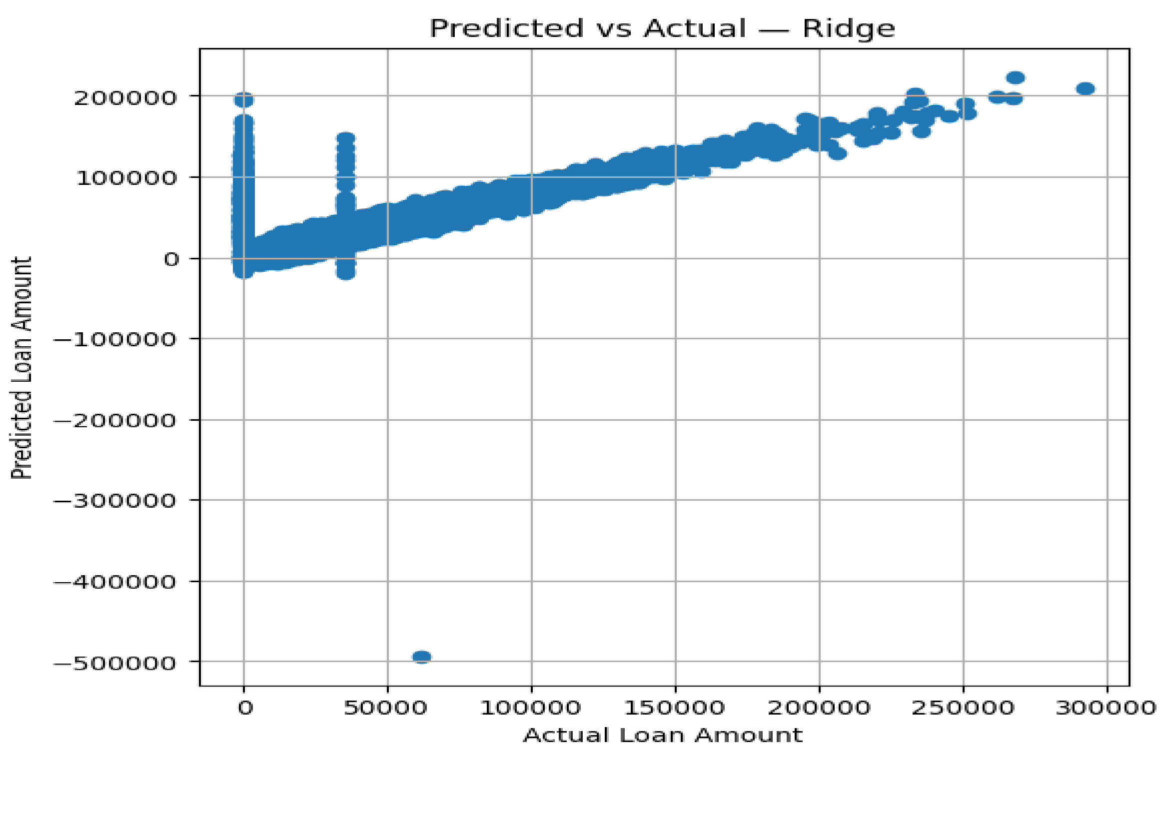


Figure 11: Predicted vs actual values plot - Lasso

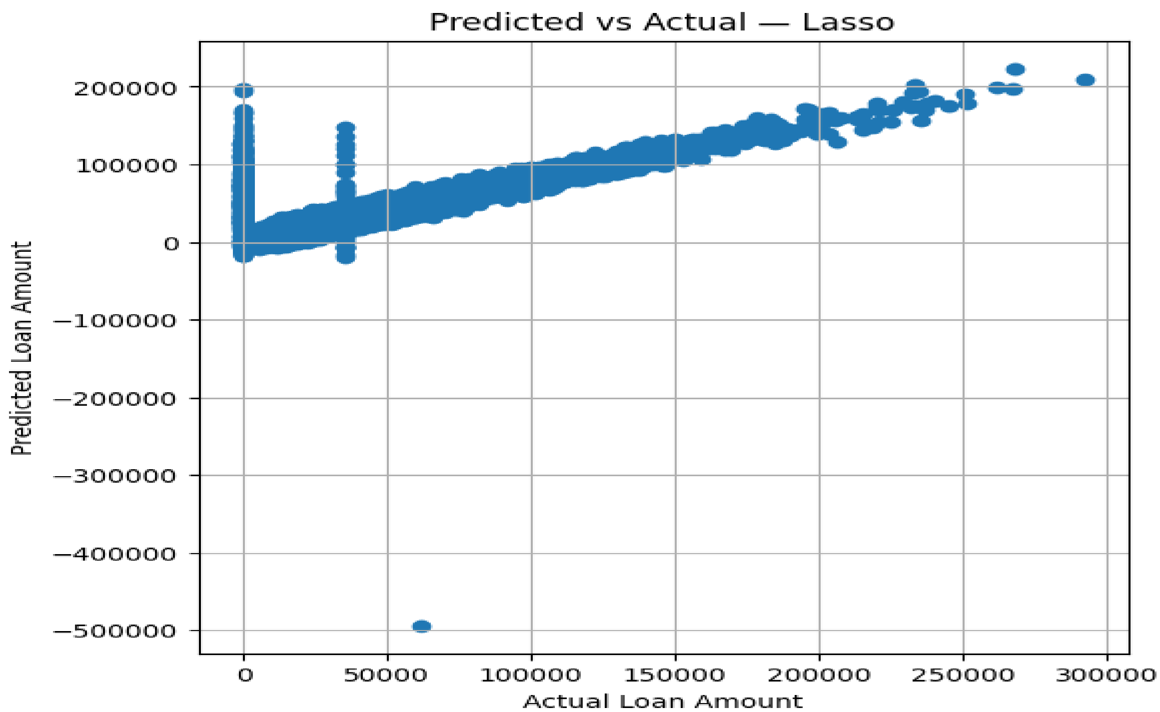
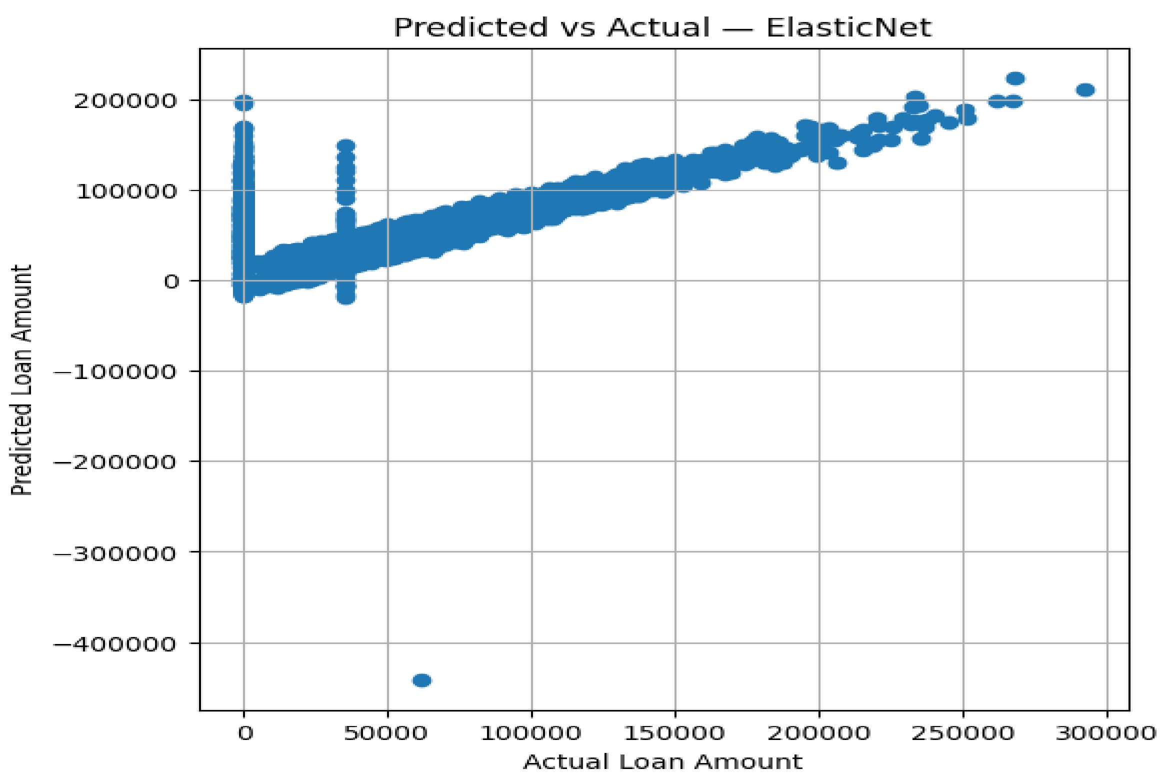


Figure 12: Predicted vs actual values plot - ElasticNet



5.4.Residual plots

Figure 13: Residual plot - Linear

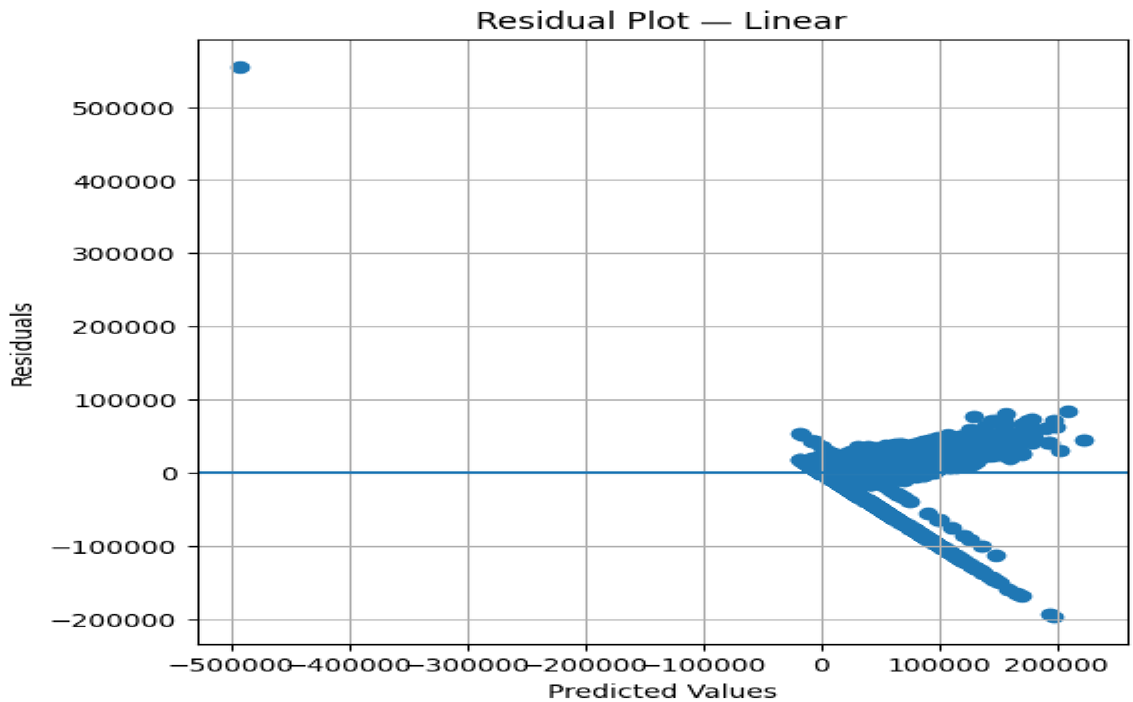


Figure 14: Residual plot - Ridge

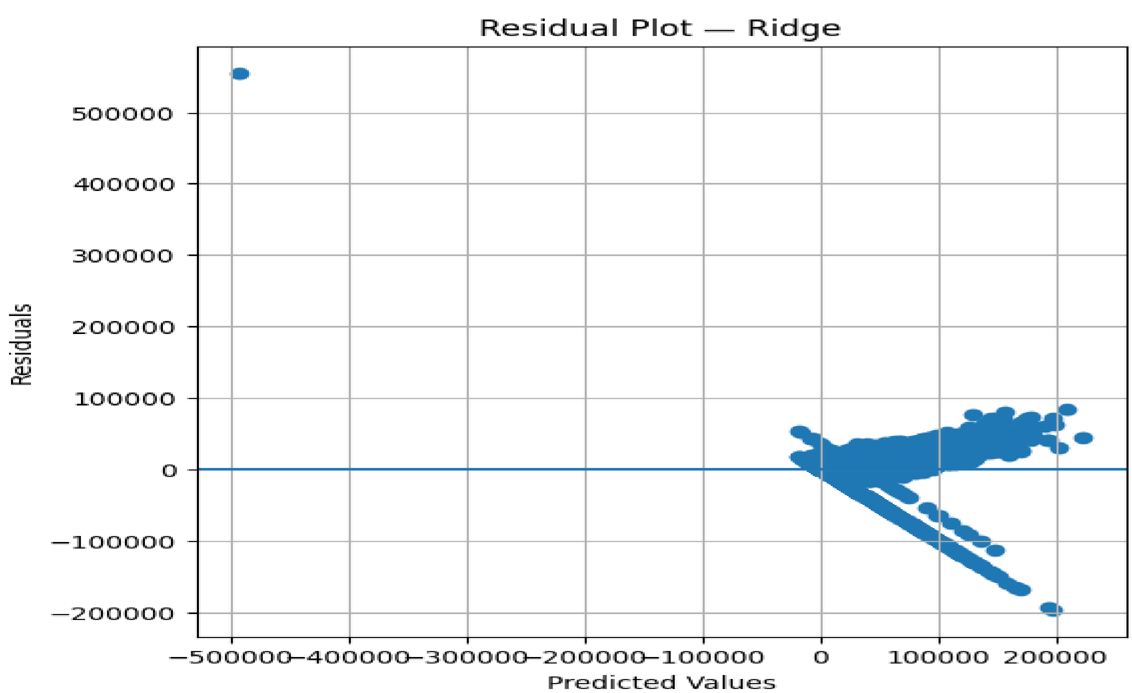


Figure 15: Residual plot - Lasso

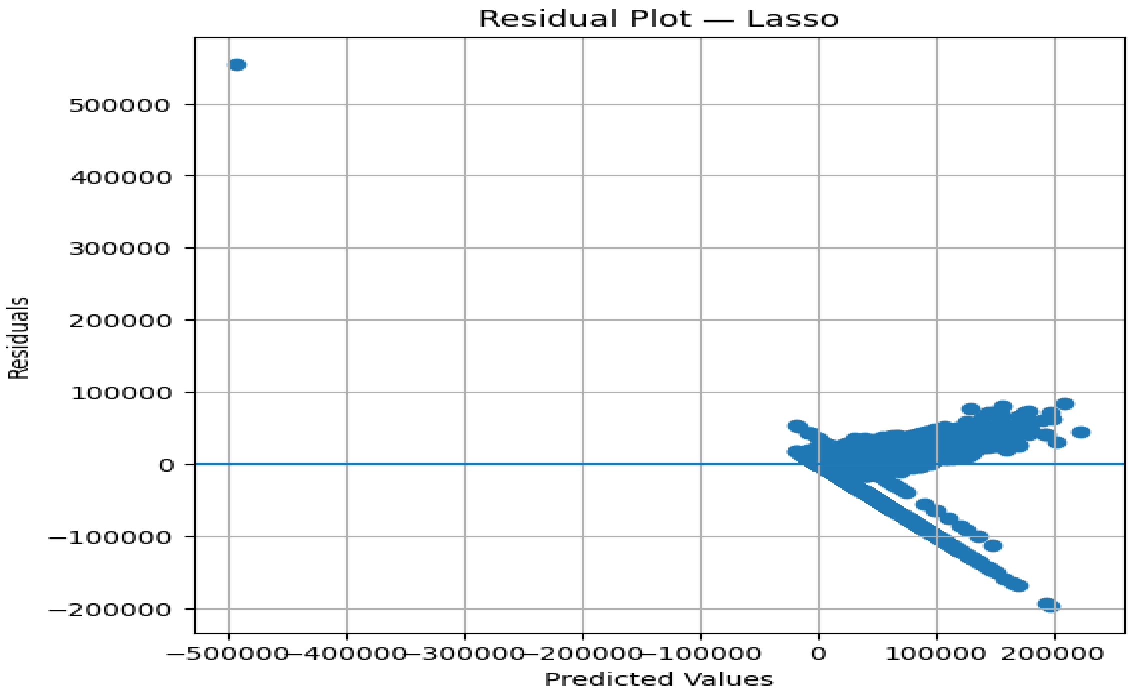
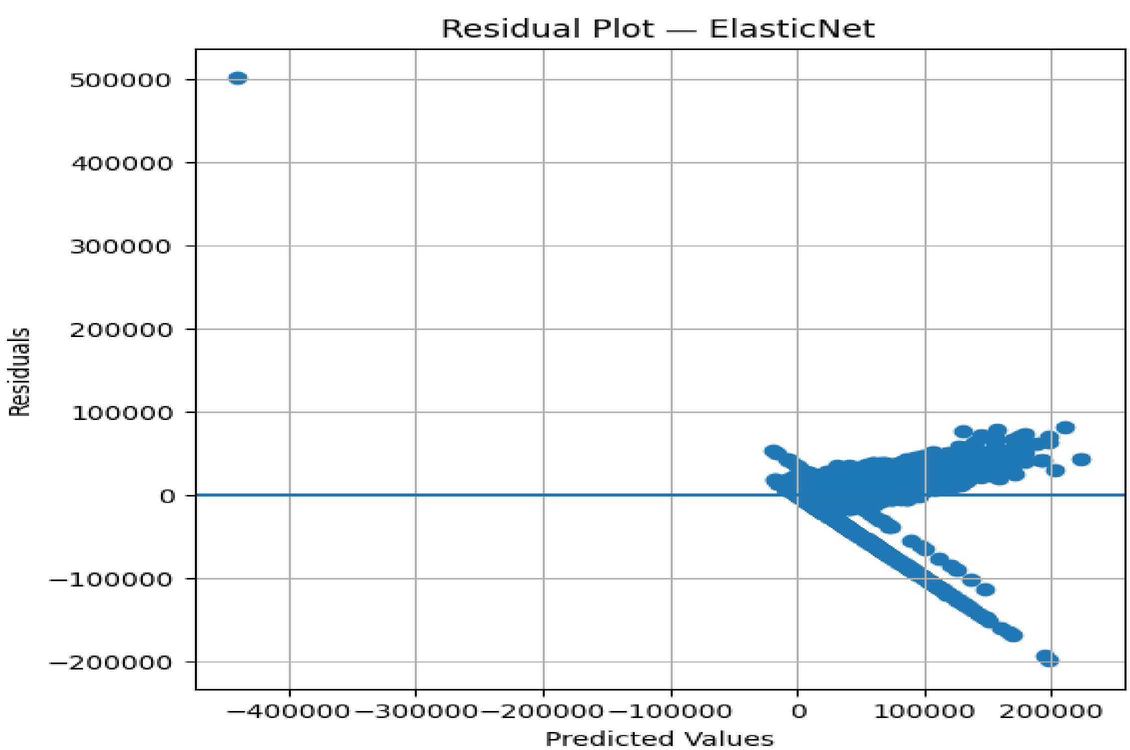


Figure 16: Residual plot – Elastic Net



5.5.Learning curves

Figure 17: Learning Curve – Linear

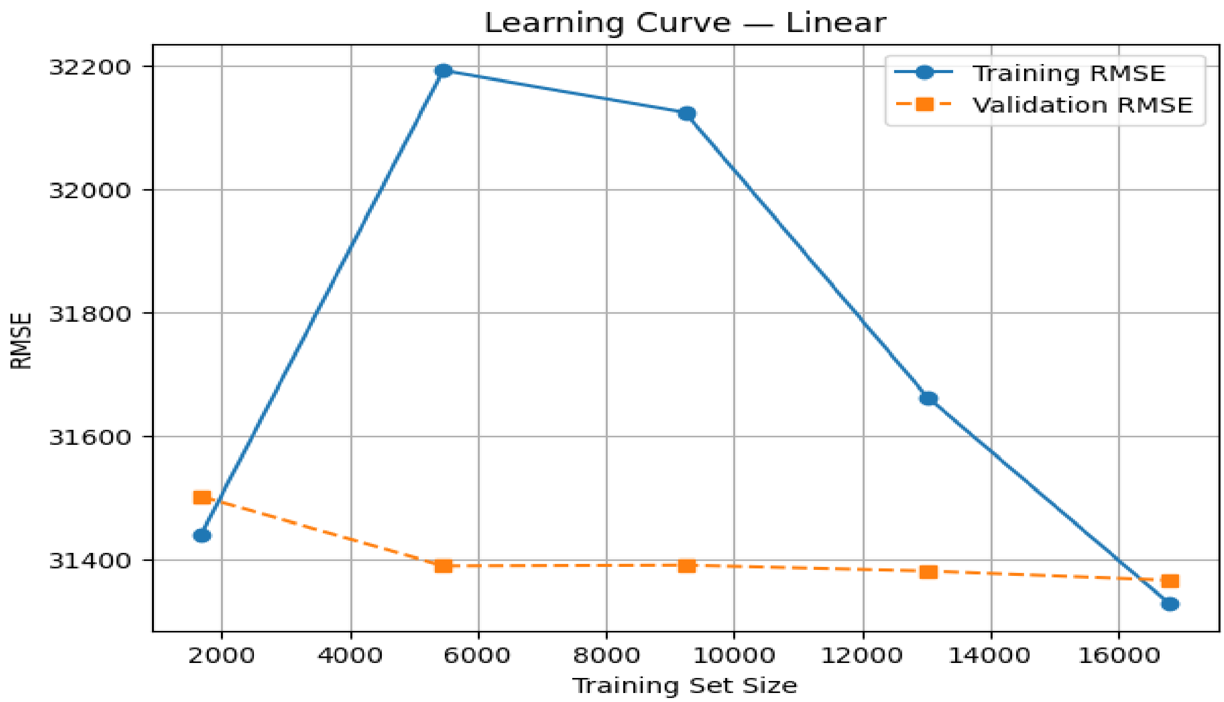


Figure 18: Learning Curve – Ridge

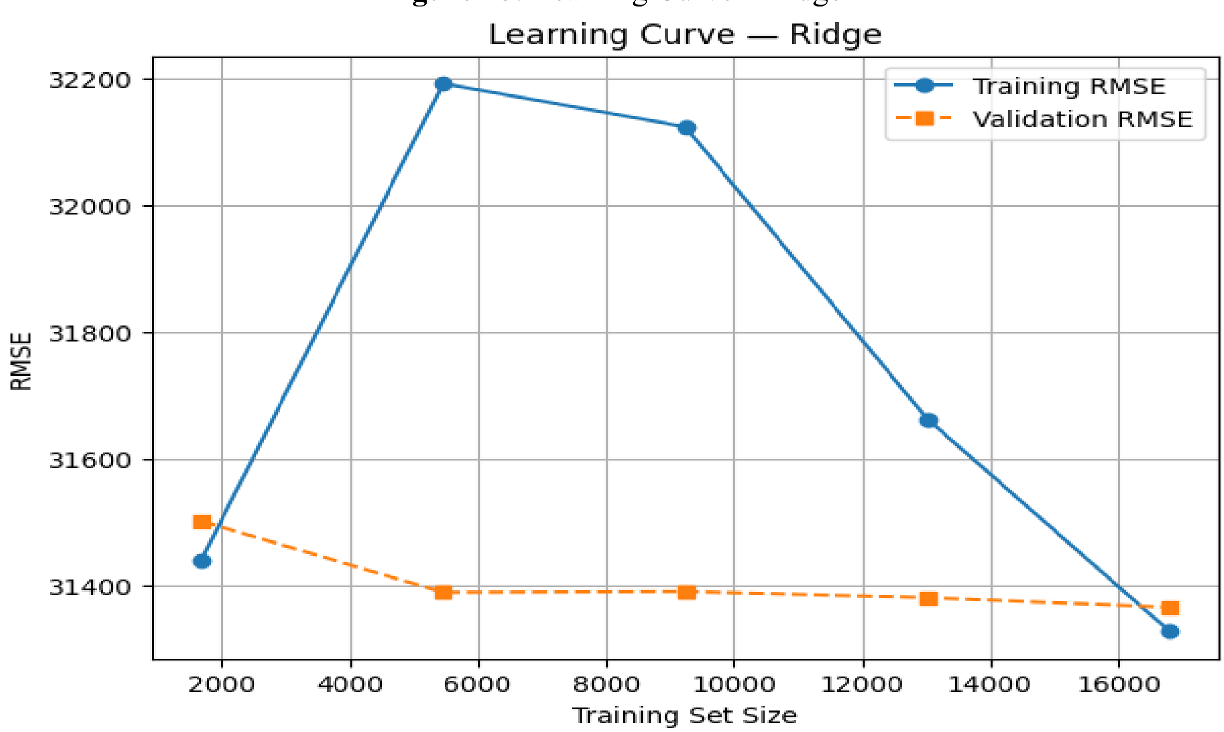


Figure 19: Learning Curve – Lasso

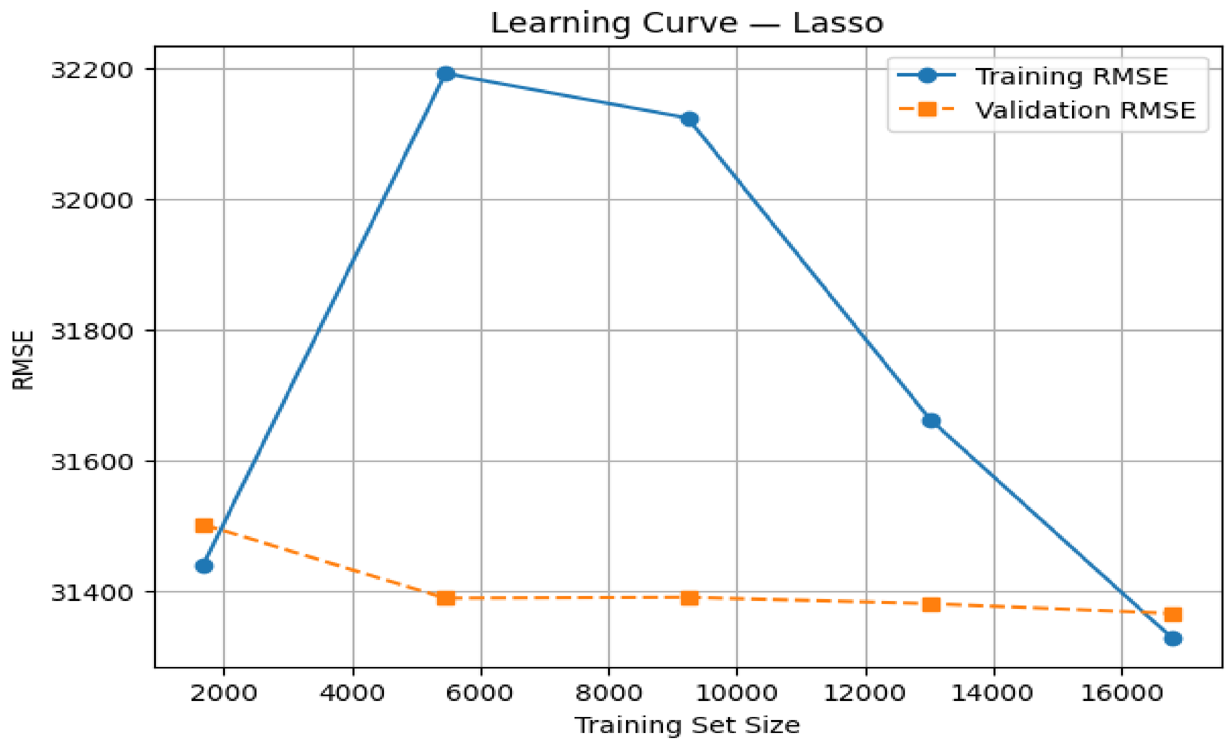
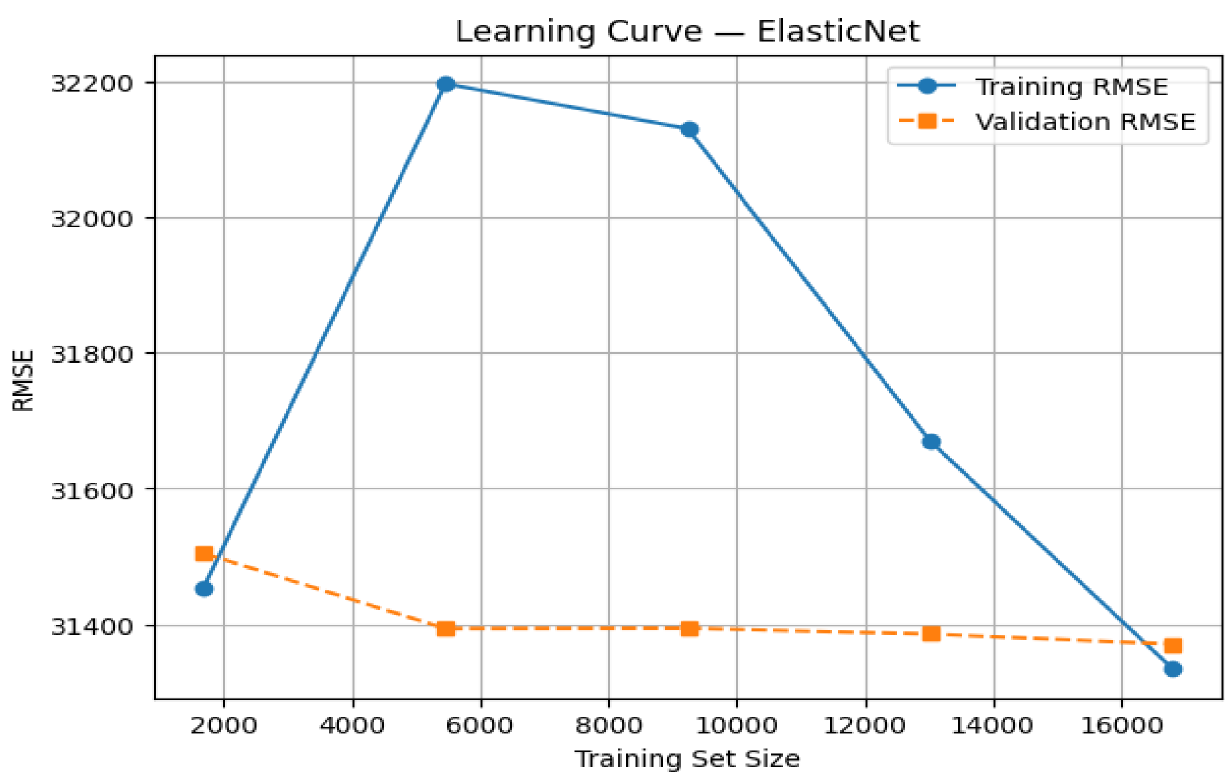
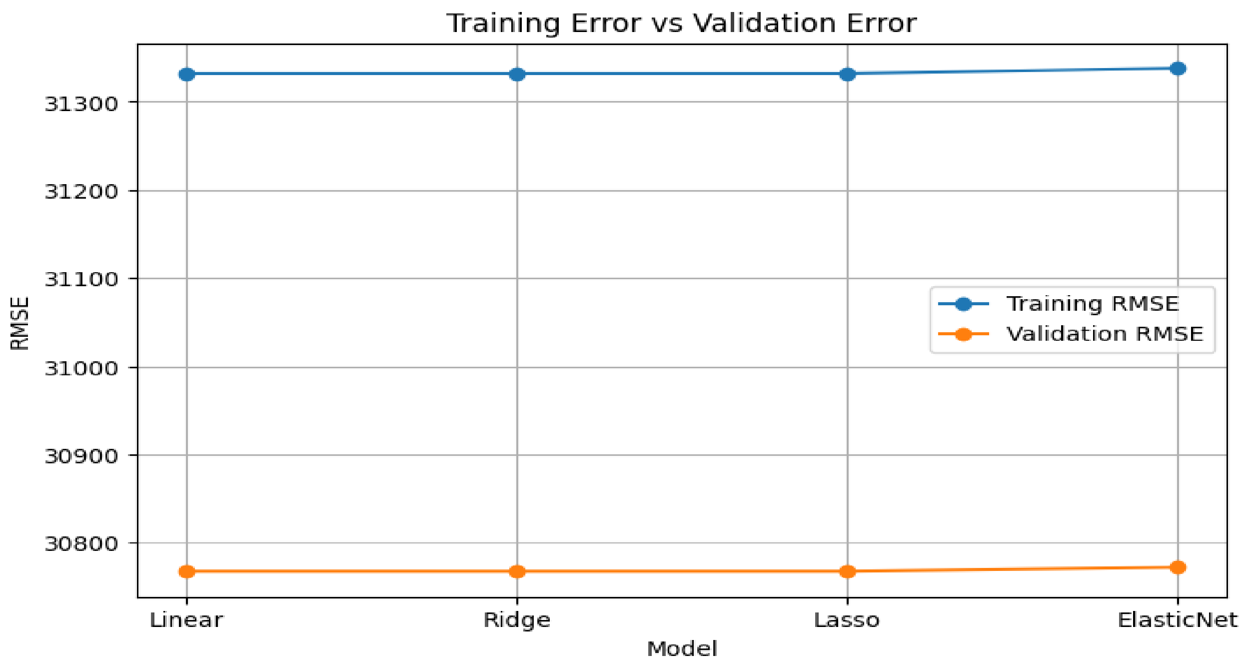


Figure 20: Learning Curve - ElasticNet



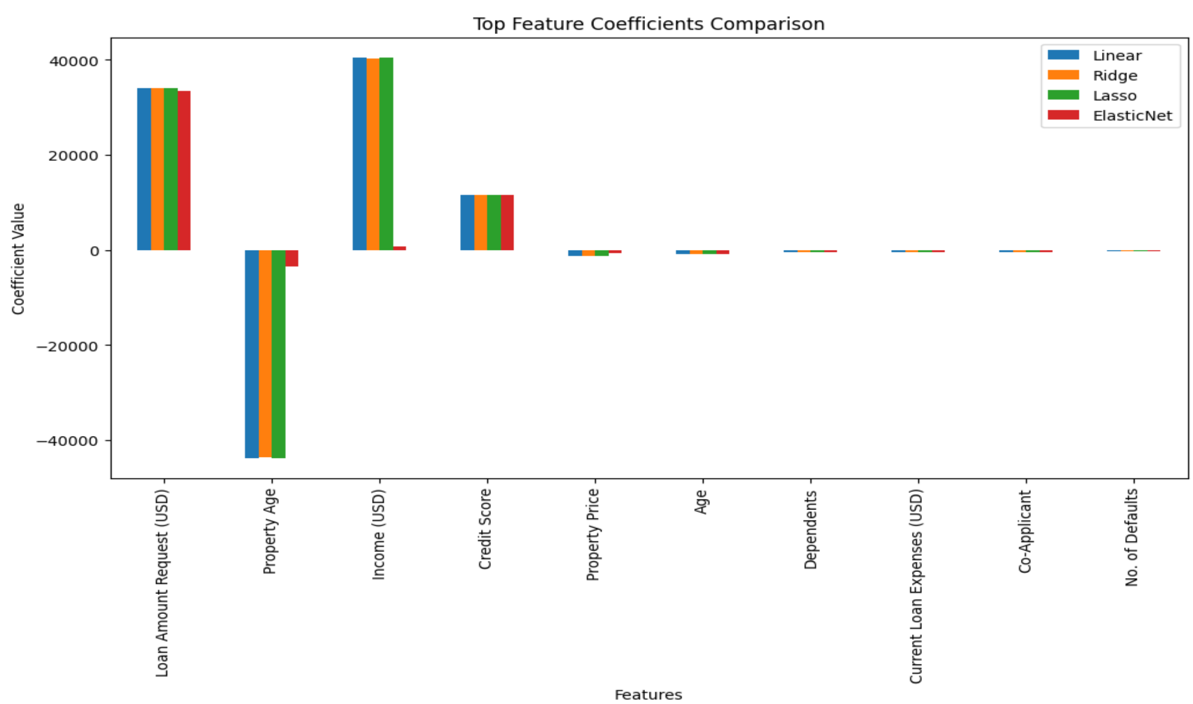
5.6.Training error vs validation error plots

Figure 21: Training error vs validation error plot



5.7. Coefficient comparison bar plots

Figure 22: Top features Coefficient Comparison



6. Performance Tables:

Hyperparameter Tuning Results:

Table 1: Hyperparameter Tuning Summary

Model	Search Method	Best Parameters	Best CV R^2
Ridge Regression	Grid Search	$\alpha = 0.01$	0.5756
Lasso Regression	Grid Search	$\alpha = 0.001$	0.5756
Elastic Net Regression	Grid Search	$\alpha = 0.01$, l1_ratio = 0.8	0.5755

Cross-Validation Performance (K = 5)

Table 2: Cross-Validation Performance

Model	MAE	MSE	RMSE	R^2
Linear Regression	21608.39	946640068.26	30767.51	0.5820
Ridge Regression	21608.32	946638464.09	30767.49	0.5820
Lasso Regression	21608.39	946639980.22	30767.51	0.5820
Elastic Net Regression	21617.67	946917076.17	30772.01	0.5819

Test Set Performance

Table 3: Test Set Performance

Model	MAE	MSE	RMSE	R^2
Linear Regression	22016.23	1033861469.17	32153.72	0.5416
Ridge Regression	22016.19	1033835237.94	32153.31	0.5416
Lasso Regression	22016.22	1033860289.04	32153.70	0.5416
Elastic Net Regression	22015.45	1021966306.24	31968.21	0.5469

Effect of Regularization on Coefficients

Table 4: Coefficient Comparison

Feature	Linear	Ridge	Lasso	Elastic Net
Loan Amount Request (USD)	34102.02	34101.90	34102.00	33393.48
Income (USD)	40575.33	40383.02	40565.75	654.43
Credit Score	11685.69	11685.66	11685.69	11666.61

7. Overfitting and Underfitting Analysis

- **Difference between training and validation errors:**
Linear Regression showed a noticeable gap between training and validation errors, indicating higher variance. Ridge and Elastic Net reduced this gap, showing improved generalization, while Lasso sometimes increased validation error due to higher bias.

- **Effect of regularization strength:**
Increasing regularization strength shrank coefficients and reduced variance, whereas very large values caused underfitting. Moderate values provided the best trade-off between complexity and accuracy.

- **Improvement in generalization after tuning:**
Cross-validated hyperparameter tuning improved validation and test performance for the regularized models, with Elastic Net giving the most balanced results.

8. Bias–Variance Analysis

• Bias behavior of Linear Regression:
Linear Regression exhibited relatively low bias but higher variance.

• Variance reduction using Ridge and Elastic Net:
Ridge and Elastic Net reduced variance through coefficient shrinkage and produced more stable predictions.

• Feature sparsity effect in Lasso:
Lasso forced several feature coefficients to zero, improving interpretability but slightly increasing bias.

9. Observations and Conclusion:

The experimental results showed that regularized regression models performed better than the baseline Linear Regression on unseen data. Ridge Regression effectively reduced coefficient magnitudes and handled multicollinearity, while Lasso Regression produced sparse and interpretable models by eliminating weak features. Elastic Net achieved the best overall balance between prediction accuracy and model complexity after hyperparameter tuning. Comparisons of training and validation errors and analysis through learning curves confirmed improved generalization and reduced overfitting for the tuned models. Overall, the experiment highlights the importance of regularization and systematic model evaluation in regression tasks, with Elastic Net identified as the most suitable model for predicting loan sanction amounts.